Alim Yakub

Research Associate | Energy & Sustainability

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https://bukaymila.github.io/home Alim BMY

Skills

Python
NumPy, Pandas, Scikit-learn, Matplotlib, SciPy

Python machine learning
Regression, Decision Trees, Controller

3D CAD modelling
Autodesk Inventor, SolidWorks, Autodesk REVIT, IESVE

Data analytics and visualization
HTML/JavaScript, Plotly, D3

Engineering & Human-centered design
HMI, Sustainable design

Research Interests

Scientific and Social Aspects of Sustainability, Distributed Energy Resources, Design, Renewable Energy Generation, Robotics & Controls, Energy Storage

Languages

English

Mandarin

Malay

Publications

Decision Tree Variations and Online Tuning for Real-Time Control of a Building in a Two-Stage Management Strategy
DOI: 10.3390/en17112730

References

Rémy Rigo-Mariani
Research Scientist
remy.rigo-mariani@g2elab.grenoble-inp.fr
Centre national de la recherche scientifique (CNRS)

Michael Lau
Associate Professor
michael.lau@ncl.ac.uk
Newcastle University

Vanessa Evers
Director
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NTU Institute of Science and Technology for Humanity

Summary

Research Engineer bridging sustainable thermal design and data science. Currently developing low-cost cold chain solutions via 3D prototyping, PCM experimentation, and Python analytics. Experienced in ML optimization, experimental design, and academic publishing across energy systems and engineering research.

Education

Newcastle University — MSc Energy & Sustainability • Distinction
Engineering — Singapore • 2018-2019

Newcastle University — BSc MDME • Hons (2:1)
Engineering — Singapore (SIT) • 2016-2018
Mechanical Design & Manufacturing Engineering

Experience

Republic Polytechnic — Singapore
Research Staff — July 2025 - Present

Designed and iterated cold box prototypes via CAD and FDM 3D printing for rapid design-test cycles. Conducted material trade-offs to select sustainable, low-cost insulation and structural materials.

Designed and executed DOE (Design of Experiments) frameworks for shelf-life stability and PCM thermal characterization. Developed SOPs for all experimental protocols, ensuring reproducibility and lab safety.

Built HTML/JavaScript and Python dashboards for data analysis/visualization.

Supervised several Applied Science FYP teams for shelf-life studies (microbial, vitamin A&C, texture analysis, moisture analysis and water activity). Supervised Engineering FYP students for the design, material and in-house PCM testing for cold box prototype.

CNRS@CREATE — Singapore
Research Associate — Jan 2023 - Sep 2023

Conversion of a MATLAB optimization program into Python for the purpose of replacing an optimization controller of a grid connected distributed energy system into an adaptive data-driven (machine learning) lightweight compact linear tree controller

Surbana Jurong-NTU Corporate Lab — Singapore
Research Associate — Apr 2022 - Sep 2022

Created a Python program for the generation and prediction of efficiency values for cooling systems using python machine learning regression algorithm

Created a Python program for the optimization of chiller systems sizing and daily operations using python

NTU Institute of Science & Technology for Humanity — Singapore
Research Associate — Nov 2020 - Apr 2022

Using python/JavaScript for data collection, analysis & generating data visualization of a network of NISTH fellow's publications in relation to the UN Sustainable development Goals on an online platform.

Building & design of NISTH 2021 Annual web report

Assist in report writing & idea generation for new research topics related to science & technology with humanities angles

Supervise FYP undergraduate on their social robot for elderly to promote inter-generational connectivity